

Henry Chen

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<https://github.com/Chenry513>

EDUCATION

Master of Science in Applied Computing (CGPA: 91%) British Columbia Institute of Technology	Sept. 2025 – Dec. 2026 (est.) Burnaby, BC
Bachelor of Science in Computing Science Simon Fraser University	Sept. 2020 – Aug. 2025 Burnaby, BC

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript, TypeScript

Frameworks & Technologies: Node.js, Express, REST APIs, Three.js, Pandas, NumPy, OpenCV

Databases & Tools: MongoDB, MySQL, PostgreSQL, SQLite, Git, GitHub Actions, Docker, Docker Compose, Linux, npm

WORK EXPERIENCE

Mitacs Research Intern – Robotics & Computer Vision BCIT — Seaspan Shipyards Collaboration (Supervisor: Mathew Smith)	Jan. 2026 – Present Vancouver, BC
- Developing a computer vision pipeline to reconstruct weld joints in 3D from multi-view camera imagery, producing geometric coordinate outputs to support robotic welding path planning. - Built WeldPath Viz, a browser-based viewer enabling engineers to explore reconstructed models, define weld start and end points, and extract coordinates for robotic execution chenry513.github.io/weldpath-viz/visualize_weld.html. - Collaborating with robotics and mechanical engineers to interpret workflow requirements and iteratively validate that reconstruction outputs align with downstream automation needs.	
Research Assistant (Backend & Automation Engineer) Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent) dl-tools.onrender.com	May – Aug. 2025 Burnaby, BC
- Designed and implemented a GitHub-integrated backend ingestion pipeline to automate validation and integration of structured user submissions, reducing manual Django-admin update time by 60%. - Implemented a Python-based linter using schema validation to enforce Django model constraints. - Integrated GitHub Actions CI to automatically validate and process submissions on each commit. - Prevented invalid entries from being deployed by enforcing schema compliance before publication.	
Research Assistant (Backend Engineer) Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent)	Aug. – Dec. 2024 Burnaby, BC
- Engineered backend components for a Django-based research platform supporting structured LLM metadata and user-facing functionality. - Designed relational models and schema logic for structured metadata storage and preference management. - Built 8+ RESTful API endpoints using Django REST Framework for authentication and data retrieval. - Enhanced Django Admin and managed migrations to maintain data integrity across updates.	

PROJECTS

Applied Data Science, BCIT 30-Day Hospital Readmission Prediction	Dec. 2025 github.com/Chenry513/COMP-9170-Final-Project
- Developed a patient-level 30-day hospital readmission risk model using 100K+ clinical encounters while preventing data leakage through GroupKFold cross-validation. - Built a reproducible ML pipeline with imbalance handling and leakage-free stacking for reliable model evaluation. - Evaluated performance using ROC-AUC, PR-AUC, and bootstrap confidence intervals to quantify uncertainty and prevent misleading performance claims in a healthcare setting.	
Affective Computing, SFU Emotion-Based Break Recommendation System	Apr. 2025 github.com/Chenry513/CMPT419_Final_project

- Built and evaluated two temporal deep learning pipelines (OpenFace+LSTM and CNN+LSTM) to detect frustration from gameplay video.
- Modeled emotional dynamics over time rather than relying on frame-level classification.
- Collected and manually annotated video clips (Anger vs Neutral), achieving 70% accuracy and $F1 = 0.70$.
- Deployed the trained model via a Flask web application enabling real-time video uploads and automated break recommendations.

Real-Time Web Application Personal Project
Rankr

July 2024
github.com/Chenry513/rankr

- Built a real-time polling system enabling groups to create, join, and rank shared options collaboratively.
- Implemented live updates using React, TypeScript, Socket.io, and Redis for low-latency synchronization.
- Designed backend logic to aggregate ranked votes and compute final group decisions.

Computer Vision, SFU
CIFAR-100 Image Classification

May 2023
github.com/Chenry513/PyTorch-Image-Classifier

- Designed and trained a convolutional neural network on the CIFAR-100 dataset (100 classes, 60K images), achieving competitive validation performance.
- Ranked 13th out of 120 students in a graduate-level computer vision Kaggle competition.
- Improved generalization using data augmentation and regularization to reduce overfitting.